

## Normalisation

### ReadMore Community Library Catalogue Search Output

#### UNF

CATALOGUE\_ENTRY(catalogue\_call\_no, catalogue\_title, (author\_id, author\_fname, author\_lname), (subject\_id, subject\_desc), catalogue\_contents, (publisher\_id, publisher\_name), catalogue\_pub\_year, catalogue\_ed, catalogue\_no\_pages, catalogue\_lang, catalogue\_note, (book\_isbn\_no, book\_type))

#### 1NF

CATALOGUE\_ENTRY(catalogue\_call\_no, catalogue\_title, catalogue\_contents, catalogue\_pub\_year, catalogue\_ed, catalogue\_no\_pages, catalogue\_lang, catalogue\_note)

**CK:** catalogue\_call\_no

CATALOGUE\_AUTHOR(catalogue\_call\_no, author\_id, author\_fname, author\_lname)

**CK:** catalogue\_call\_no, author\_id

CATALOGUE\_SUBJECT(catalogue\_call\_no, subject\_id, subject\_desc)

**CK:** catalogue\_call\_no, subject\_id

CATALOGUE\_PUBLISHER(catalogue\_call\_no, publisher\_id, publisher\_name)

**CK:** catalogue\_call\_no, publisher\_id

BOOK(book\_isbn\_no, book\_type, catalogue\_call\_no)

**CK:** book\_isbn\_no

#### Partial dependencies:

author\_id --> author\_fname, author\_lname

subject\_id --> subject\_desc

publisher\_id --> publisher\_name

## 2NF

CATALOGUE\_ENTRY(catalogue\_call\_no, catalogue\_title, catalogue\_contents, catalogue\_pub\_year, catalogue\_ed, catalogue\_no\_pages, catalogue\_lang, catalogue\_note)

CATALOGUE\_AUTHOR(catalogue\_call\_no, author\_id)

AUTHOR(author\_id, author\_fname, author\_lname)

CATALOGUE\_SUBJECT(catalogue\_call\_no, subject\_id)

SUBJECT(subject\_id, subject\_desc)

CATALOGUE\_PUBLISHER(catalogue\_call\_no, publisher\_id)

PUBLISHER(publisher\_id, publisher\_name)

BOOK(book\_isbn\_no, book\_type, catalogue\_call\_no)

**Transitive dependencies:** N/A

## 3NF

CATALOGUE\_ENTRY(catalogue\_call\_no, catalogue\_title, catalogue\_contents, catalogue\_pub\_year, catalogue\_ed, catalogue\_no\_pages, catalogue\_lang, catalogue\_note)

CATALOGUE\_AUTHOR(catalogue\_call\_no, author\_id)

AUTHOR(author\_id, author\_fname, author\_lname)

CATALOGUE\_SUBJECT(catalogue\_call\_no, subject\_id)

SUBJECT(subject\_id, subject\_desc)

CATALOGUE\_PUBLISHER(catalogue\_call\_no, publisher\_id)

PUBLISHER(publisher\_id, publisher\_name)

BOOK(book\_isbn\_no, book\_type, catalogue\_call\_no)

**Full dependencies:**

catalogue\_call\_no --> catalogue\_title, catalogue\_contents, catalogue\_pub\_year,

catalogue\_ed, catalogue\_no\_pages, catalogue\_lang, catalogue\_note

author\_id --> author\_fname, author\_lname

subject\_id --> subject\_desc

publisher\_id --> publisher\_name

book\_isbn\_no --> book\_type, catalogue\_call\_no

## ReadMore Community Library Borrowers Quarterly Report

### UNF

BORROWER(borrower\_no, borrower\_fname, borrower\_lname, borrower\_class, borrower\_status, (branch\_code, book\_copy\_id, branch\_name, loan\_current\_date\_time, loan\_due\_date, loan\_actual\_return\_date\_time, loan\_fine\_amount, loan\_fine\_paid\_date))

### 1NF

BORROWER(borrower\_no, borrower\_fname, borrower\_lname, borrower\_class, borrower\_status)

**CK:** borrower\_no

LOAN (branch\_code, book\_copy\_id, loan\_current\_date\_time, branch\_name, loan\_due\_date, loan\_actual\_return\_date\_time, loan\_fine\_amount, loan\_fine\_paid\_date, borrower\_no)

**CK:** branch\_code, book\_copy\_id, loan\_current\_date\_time

### Partial Dependency:

branch\_code --> branch\_name

### 2NF

BORROWER(borrower\_no, borrower\_fname, borrower\_lname, borrower\_class, borrower\_status)

LOAN (branch\_code, book\_copy\_id, loan\_current\_date\_time, loan\_due\_date, loan\_actual\_return\_date\_time, loan\_fine\_amount, loan\_fine\_paid\_date, borrower\_no)

BRANCH (branch\_code, branch\_name)

**Transitive Dependency:** N/A

### 3NF

BORROWER (borrower\_no, borrower\_fname, borrower\_lname, borrower\_class, borrower\_status)

LOAN (branch\_code, book\_copy\_id, loan\_current\_date\_time, loan\_due\_date, loan\_actual\_return\_date\_time, loan\_fine\_amount, loan\_fine\_paid\_date, borrower\_no)

BRANCH (branch\_code, branch\_name)

**Full Dependency:**

borrower\_no --> borrower\_fname, borrower\_lname, borrower\_class, borrower\_status)

branch\_code, book\_copy\_id, loan\_current\_date\_time --> loan\_due\_date,  
loan\_actual\_return\_date\_time, loan\_fine\_amount, loan\_fine\_paid\_date, borrower\_no

branch\_code --> branch\_name

**Final relations**

BORROWER (borrower\_no, borrower\_fname, borrower\_lname, borrower\_class,  
borrower\_status)

LOAN (branch\_code, book\_copy\_id, loan\_current\_date\_time, loan\_due\_date,  
loan\_actual\_return\_date\_time, loan\_fine\_amount, loan\_fine\_paid\_date, borrower\_no)

BRANCH (branch\_code, branch\_name)

CATALOGUE\_ENTRY(catalogue\_call\_no, catalogue\_title, catalogue\_contents,  
catalogue\_pub\_year, catalogue\_ed, catalogue\_no\_pages, catalogue\_lang,  
catalogue\_note)

CATALOGUE\_AUTHOR(catalogue\_call\_no, author\_id)

AUTHOR(author\_id, author\_fname, author\_lname)

CATALOGUE\_SUBJECT(catalogue\_call\_no, subject\_id)

SUBJECT(subject\_id, subject\_desc)

CATALOGUE\_PUBLISHER(catalogue\_call\_no, publisher\_id)

PUBLISHER(publisher\_id, publisher\_name)

BOOK(book\_isbn\_no, book\_type, catalogue\_call\_no)